

OPERATING SYSTEMS

Storage Management

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The operating system is responsible for using hardware efficiently.

For the disk drives, this means having a fast *access time & disk bandwidth*.

Access time has three major components:

Seek time is the time for the disk to **move the heads** to the cylinder containing the desired sector

Rotational latency time waiting for the disk to rotate the desired sector to the disk head

Transfer rate is the time taken to transfer the bytes from the disk

Disk access time = seek time + rotational latency + transfer time

Disk seek time = No of cylinders * time taken for moving 1 cylinder

Disk bandwidth is the total number of bytes transferred *divided* by the total time between the first request for service and the completion of the last transfer.

Several algorithms exist to schedule the servicing of disk I/O requests

Scheduling Algorithms: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK

First request is served First

No need to order the request

Serve as it comes

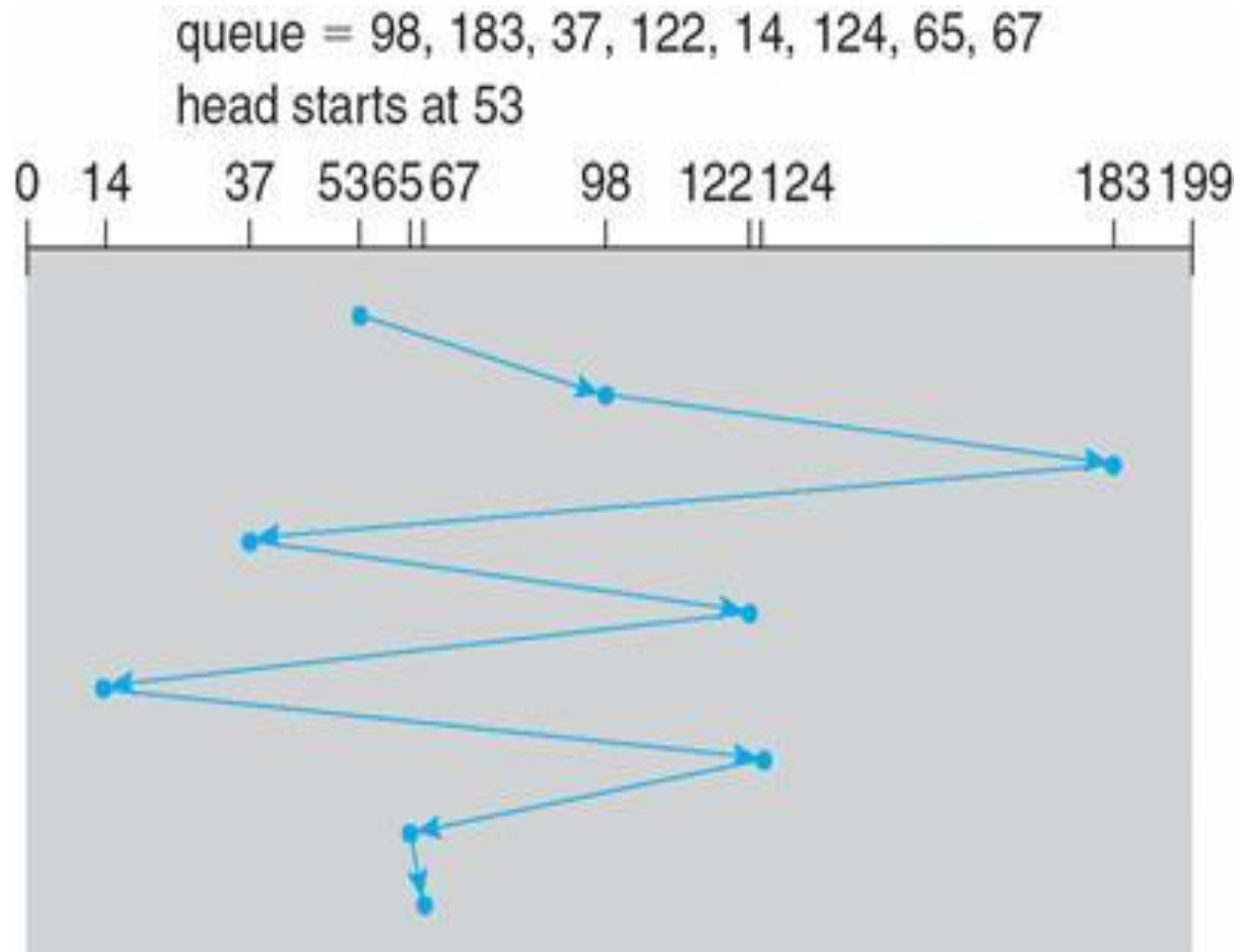
We illustrate them with a **Request Queue** (cylinder range 0-199):

98, 183, 37, 122, 14, 124, 65, 67

Head pointer: cylinder 53 (current position)

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Disk Scheduling Algorithm - FCFS



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Disk Scheduling Algorithm - FCFS

$$\begin{aligned}\text{Total Head movement} &= |53-98| + |98-183| + \\ &\quad |183-37| + |37-122| + \\ &\quad |122-14| + |14-124| + \\ &\quad |124-65| + |65-67| \\ &= 45 + 85 + 146 + 85 + 108 + 110 + 59 + 2 \\ &= 640 \text{ Cylinders}\end{aligned}$$

Head movement shows 640 cylinders

Advantages

- Every request gets a fair chance
- No indefinite postponement

Disadvantages

- Does not try to optimize seek time
- May not provide the best possible service

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Disk Scheduling Algorithm - SSTF

Selects the request with the **minimum seek time** from the current head position

SSTF scheduling may cause **starvation** of some requests

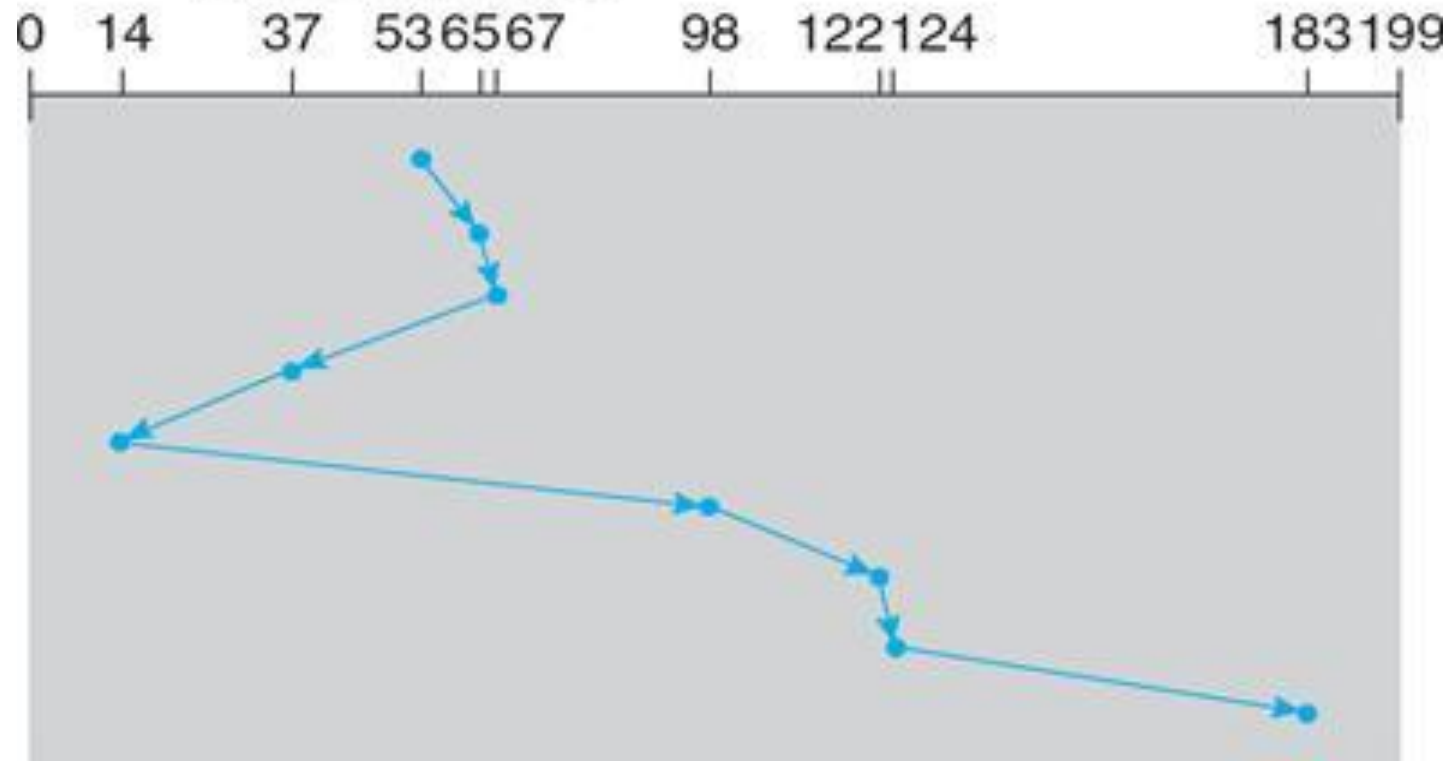


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Disk Scheduling Algorithm - SSTF

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



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Disk Scheduling Algorithm - SSTF

$$\begin{aligned}\text{Total Head movement} &= |53-65| + |65-67| + \\ &\quad |67-37| + |37-14| + \\ &\quad |14-98| + |98-122| + \\ &\quad |122-124| + |124-183| \\ &= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 \\ &= 236\end{aligned}$$

Head movement shows 236 cylinders

Advantages

- Average Response Time decreases
- Throughput increases

Disadvantages

- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has higher seek time as compared to incoming requests
- High variance of response time as SSTF favours only some requests



THANK YOU

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